

3. REQUIREMENT ANALYSIS

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Date	01 October 2026
Team ID	SWTID-2026-4302
Project Name	Import Data using Transform Maps (Spreadsheet)
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3.1 Customer Journey Map

A customer journey map shows the stages a user goes through to reach a goal, with their interactions, motivations, pain points and opportunities. The journey below follows a ServiceNow administrator working with HR, from preparing the spreadsheet to reporting on the imported data (reference: Nielsen Norman Group, “Journey Mapping 101”).

Scenario: preparing, importing, updating and reporting on employee data using Import Sets and Transform Maps. **Persona:** ServiceNow administrator working with HR. The map is based on the project scenario and the team’s assumptions rather than customer interviews.

How to read the map (next page)

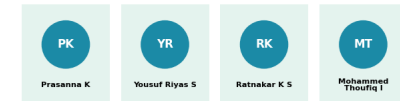
- **Stages (columns):** Prepare → Set up & load → Map & transform → Update (Coalesce) → Report, covering 12 steps in total.
- **Rows:** Steps, Interactions (touchpoints), Goals & motivations (“Help me...”), Positive moments, Negative moments and Areas of opportunity.
- **Sticky colours:** purple = steps, blue = interactions, orange = goals, green = positive, pink = negative, yellow = opportunities.

Key insights from the journey

- The biggest risk is in the **Update (Coalesce)** stage: a non-unique coalesce field would still create duplicates.
- Most pain points are about **setup details** – matching headers, sheet number / header row, and one-by-one field creation.
- Positive moments cluster around **visible results**: the success message, Auto Map, the Transform History counts and the dashboard.
- Opportunities feed the Future Scope: templates, validation, scheduled loads, import notifications and an import success / error report.

Employee data import

Persona and steps based on the project scenario (team assumptions)



SCENARIO Preparing, importing, updating and reporting on employee data from a spreadsheet													
1 Prepare		2 Set up & load				3 Map & transform			4 Update (Coalesce)		5 Report		
How does the work first begin?		What does the admin do to begin the import?				In the core moments, what happens?			What happens as the import process finishes?		What happens after the import is over?		
S	Steps	Receive employee data	Prepare Google Sheet	Download as .xlsx	Create target table	Load the data	Create Transform Map	Run the transform	Verify the records	Set Coalesce	Re-import & check history	Create reports	Dashboard & share
	What does the person typically do?	Employee data arrives from HR as an Excel / Google Sheet	Admin creates sample data: Employee ID, Name, Email, Department, Location	Download the sheet as Sample Spreadsheet.xlsx	Create Employee Test (u_employee_test) with 5 String fields	Load Data creates the Employee Import staging table (u_employee_import)	Sample Spreadsheet Import: Employee Import to Employee Test; Auto Map	Click Transform, select the map and run it	Open Employee Test and check the imported rows	Coalesce = true on the Employee ID field map	Upload the updated sheet, transform, open Transform History	Three reports on Employee Test: by Department (pie), by Location, Employee List	Employee Analytics Dashboards with the three reports, share with HR
I	Interactions	Email or shared file from HR	Google Sheets in the browser	Google Sheets > File > Download > Microsoft Excel	Tables > Create New	System Import Sets > Load Data	Transform Maps; Field Maps related list	Load Data > Run Transform link	Employee Test list	Transform Map > Field Maps > Coalesce	Load Data (existing table)	All > Reports > New	PA_DASHBOARDS FORM
	What interactions or touchpoints do they use? People - Places - Things	People: HR data owner, ServiceNow admin			Form Layout / Form Context Menu	Sheet number 1, Header row 1		Success message / state Complete	Personalize List Columns		Transform History counts	Table: Employee Test; Count	Share > Add to Dashboards
G	Goals & motivations	Help me get the employee data ready to load	Help me keep the source data clean and simple	Help me get a file ServiceNow can read	Help me have a place to store employees	Help me get the data into ServiceNow fast	Help me send each column to the right field	Help me move the staged rows into the table	Help me confirm the data is correct	Help me avoid duplicate employees	Help me trust that repeated imports are safe	Help me see employees by department and location	Help me share insights with HR
	At each step, what is the primary goal? ("Help me...")												
P	Positive moments	Sample data is simple (only 5 columns)			Fields appear on the Employee Test form	Success message after loading	Auto Map Matching Fields fills the mapping	Quick, no-code transform with a success message	All records are visible in the table	Existing rows update; new rows insert	Same sheet again: 0 inserted, 0 updated, 4 ignored	Charts show the department and location split	One dashboard gives HR a single view
	What steps are enjoyable, productive or exciting?												
N	Negative moments	Data arrives in Excel with no direct route into the Employee table		Headers must match the target field names	Fields have to be added one at a time	Sheet number and header row must be right	Un-mapped fields are not transferred		Column order needs rearranging	A non-unique coalesce field would still create duplicates			Dashboard form needs ACL Admin Overrides = true
	What steps are frustrating, confusing or time-consuming?												
A	Areas of opportunity	How might we provide a ready spreadsheet template to HR?		Add a header check before uploading	Script or reuse table definitions	Schedule recurring loads with a data source	Add validation on required fields (ID, Name, Email)		Set a default list view for the table		Email the import counts after each run	Add an import success / error report	Restrict sharing by users, groups and roles
	How might we make each step better?												

Figure 3.1 – Customer journey map: employee data import into ServiceNow (Prepare, Set up & load, Map & transform, Update (Coalesce), Report)

3.2 Solution Requirement

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Spreadsheet Preparation	Create a Google Spreadsheet with Employee ID, Employee Name, Email, Department and Location Download it as Sample Spreadsheet.xlsx
FR-2	Table Creation	Create custom table Employee Test (u_employee_test) Add five String fields through Form Layout and verify the form
FR-3	Import Set Table (Load Data)	Create Employee Import (u_employee_import) staging table from the spreadsheet Upload with Sheet number 1 and Header row 1
FR-4	Transform Map	Create Sample Spreadsheet Import (source Employee Import, target Employee Test) Auto Map Matching Fields and confirm the field maps
FR-5	Transform & Validate	Run the transform and confirm the success message Verify records and required fields (Employee ID, Name, Email) in Employee Test
FR-6	Coalesce (Duplicate Prevention)	Set Coalesce = true on Employee ID Existing IDs are updated, new IDs are inserted, unchanged rows are ignored
FR-7	Reports & Dashboard	Reports: Employees by Department (pie), Employees by Location, Employee List Dashboard: Employee Analytics Dashboards; share by users, groups and roles

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Entire flow is done through the ServiceNow UI (Load Data, Transform Map, Reports) with no code; list columns can be personalised.
NFR-2	Security	Import and configuration performed by an administrator; dashboards and reports can be restricted by users, groups and roles; creating dashboards through the backend form requires Admin Overrides on the pa_dashboards create ACL.
NFR-3	Reliability	Coalesce on a unique Employee ID keeps repeated imports consistent; Transform History records inserted, updated, ignored and skipped rows.
NFR-4	Performance	Data is staged in an import set table and transformed in one run; run time is visible in Transform History.
NFR-5	Availability	Runs on a ServiceNow Personal Developer Instance (or an instance with development access); availability follows the instance.
NFR-6	Scalability	The same staging + transform map pattern can be reused for larger files and other target tables.

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) shows how data enters and leaves the system, what changes the information, and where data is stored. The diagram below shows the spreadsheet import, the transform with coalesce, and the reporting flow.

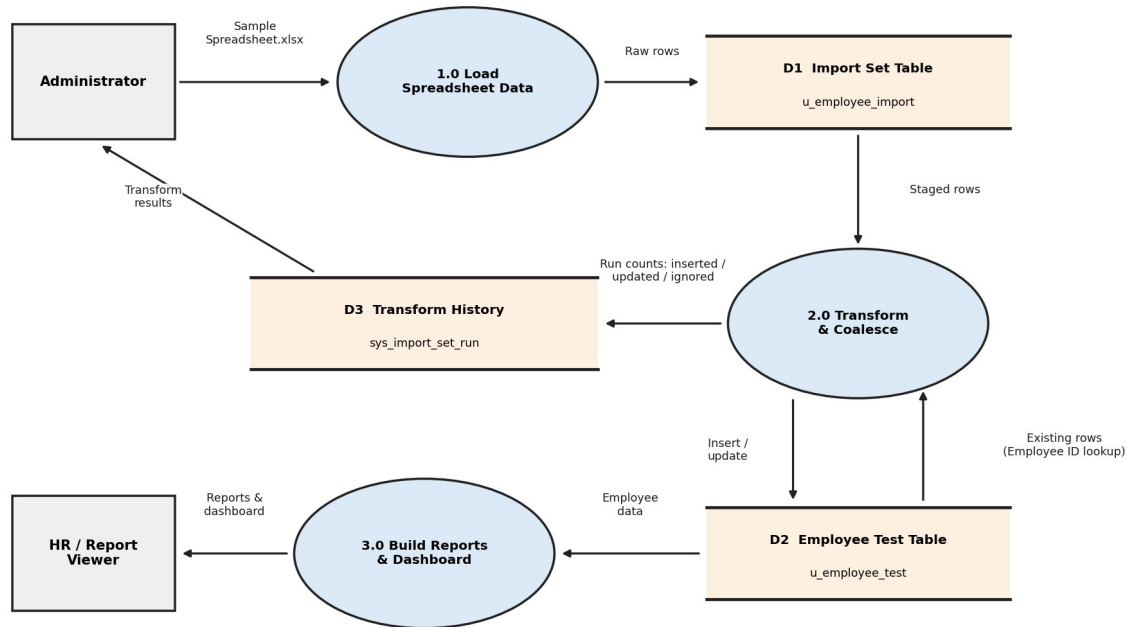


Figure 3.2 – Data flow: spreadsheet import, transform with coalesce, and reporting

Element	Name	Description
External entity	Administrator	Supplies Sample Spreadsheet.xlsx and receives transform results
Process 1.0	Load Spreadsheet Data	Reads Sheet 1 / Header row 1 and writes raw rows to the staging table
Data store D1	Import Set Table (u_employee_import)	Temporary staging area for the uploaded rows
Process 2.0	Transform & Coalesce	Maps fields, looks up existing records by Employee ID, then inserts or updates
Data store D2	Employee Test Table (u_employee_test)	Final employee records
Data store D3	Transform History (sys_import_set_run)	Run counts: inserted / updated / ignored
Process 3.0	Build Reports & Dashboard	Aggregates Employee Test data into three reports on a dashboard
External entity	HR / Report Viewer	Views and shares the reports and dashboard

User Stories

User stories list the product functionality from the user's point of view. Each story is assigned to a team member for development.

Sprint	Epic	Story / Task	Acceptance criteria	Priority	Story No.	Assigned to
Sprint-1	Spreadsheet Preparation	As an administrator, I can create a new Google Spreadsheet with sample employee data (Employee ID, Employee Name, Email, Department, Location).	Sheet contains all five columns with sample rows	High	USN-1	Yousuf Riyas S
Sprint-1		As an administrator, I can download the spreadsheet as Sample Spreadsheet.xlsx.	A .xlsx file is downloaded and opens with the same data	High	USN-2	Yousuf Riyas S
Sprint-1	Table Creation	As an administrator, I can create the custom table Employee Test (u_employee_test).	Table is listed with label Employee Test and name u_employee_test	High	USN-3	Prasanna K
Sprint-1		As an administrator, I can add Employee ID, Employee Name, Email, Department and Location as String fields using Form Layout.	All five fields are displayed on the Employee Test form	High	USN-4	Prasanna K
Sprint-2	Import Set Table	As an administrator, I can use Load Data to create the Employee Import (u_employee_import) staging table from the spreadsheet.	Success message is shown and Create Transform Map is available	High	USN-5	Ratnakar K S
Sprint-2	Transform Map	As an administrator, I can create the transform map Sample Spreadsheet Import (Employee Import to Employee Test) and auto-map matching fields.	Source and target fields are mapped in the Field Maps list	High	USN-6	Mohammed Thoufiq I
Sprint-2		As an administrator, I can run the transform and see a success message.	Transform completes successfully	High	USN-7	Ratnakar K S
Sprint-3	Transform & Validate	As an administrator, I can open the Employee Test table and verify the imported records, arranging columns with Personalize List Columns.	All spreadsheet rows appear in Employee Test	High	USN-8	Yousuf Riyas S
Sprint-3		As an administrator, I can validate that required fields (Employee ID, Name, Email) hold valid data so incorrect or incomplete rows can be corrected.	Incorrect or incomplete rows are identified for correction	Medium	USN-9	Mohammed Thoufiq I
Sprint-3	Coalesce	As an administrator, I can set Coalesce to true on the Employee ID field map.	Coalesce is true for Employee ID and the map is saved	High	USN-10	Prasanna K

Sprint	Epic	Story / Task	Acceptance criteria	Priority	Story No.	Assigned to
Sprint-3		As an administrator, I can import an updated sheet (4 rows: 2 existing IDs with changed name/email and 2 new IDs) and have existing records updated and new ones inserted.	Transform History shows 2 inserted and 2 updated	High	USN-11	Ratnakar K S
Sprint-3		As an administrator, I can re-import the same sheet without creating duplicates.	Transform History shows 0 inserted, 0 updated and 4 ignored	High	USN-12	Mohammed Thoufiq I
Sprint-4	Reports	As an HR user, I can view reports on Employee Test: Employees by Department (pie chart), Employees by Location and Employee List.	Three reports run and are saved	Medium	USN-13	Yousuf Riyas S
Sprint-4	Dashboard	As an administrator, I can create the Employee Analytics Dashboards (enabling Admin Overrides on the pa_dashboards create ACL when the form is opened by its backend name).	Dashboard form is saved	Medium	USN-14	Prasanna K
Sprint-4		As an administrator, I can add the three reports to the dashboard and restrict sharing by users, groups and roles.	Dashboard shows the three reports	Medium	USN-15	Mohammed Thoufiq I

3.4 Technology Stack

Technical architecture

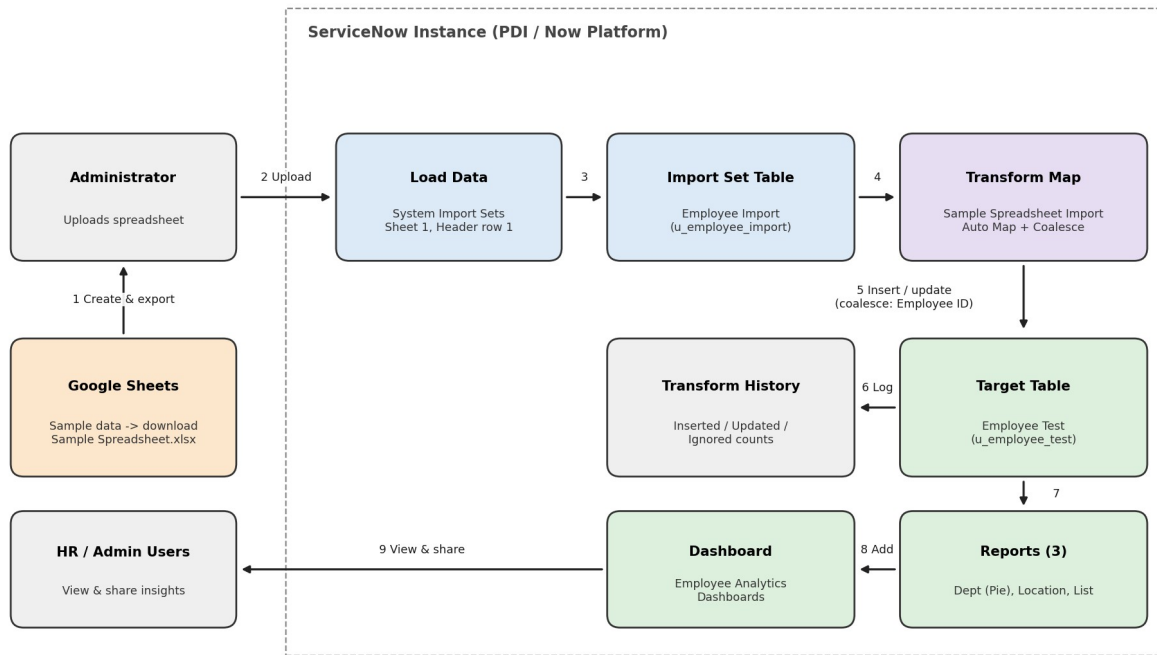


Figure 3.3 – Load Data > Import Set > Transform Map (coalesce) > Employee Test > Reports & Dashboard

Table 1 – Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Administrator and HR users work in the Application Navigator, forms, lists, Load Data page, Reports and Dashboards	ServiceNow Now Platform web UI
2	Application Logic-1	Load Data reads the spreadsheet (Sheet 1, Header row 1) into the staging table	ServiceNow Import Sets (System Import Sets)
3	Application Logic-2	Transform Map Sample Spreadsheet Import maps Employee Import fields to Employee Test fields; Coalesce on Employee ID	ServiceNow Transform Maps and Field Maps
4	Application Logic-3	Results and counts of each run (inserted, updated, ignored, skipped)	Transform History
5	Database	Staging table u_employee_import; target table u_employee_test (five String fields)	ServiceNow relational database (custom tables)
6	Cloud Database	Database service hosted with the instance	ServiceNow cloud (PDI)
7	File Storage	Source file Sample Spreadsheet.xlsx uploaded through Load Data	Google Sheets (authoring), Excel .xlsx
8	External API	Not used; import is file based	N/A
9	Machine Learning Model	Not applicable to this project	N/A
10	Infrastructure	Configured and run on an instance with development access	ServiceNow Personal Developer Instance
11	Reporting	Three reports on Employee Test combined on Employee Analytics Dashboards	ServiceNow Reports and Dashboards

Table 2 – Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	None; the solution is configured with platform-native (low-code) features	ServiceNow platform features
2	Security Implementations	Administrator-level access; ACL Admin Overrides for dashboard creation; sharing restricted by users, groups and roles	ServiceNow ACLs, roles, groups
3	Scalable Architecture	Staging table separates raw data from the target so larger files and other target tables can reuse the pattern	Import Sets + Transform Maps
4	Availability	Depends on the hosting instance; production use relies on the managed ServiceNow cloud	ServiceNow cloud instance
5	Performance	Bulk load into a staging table, single transform run, coalesce lookup on Employee ID	Import Sets, Transform Maps

References: ServiceNow Docs – Import set coalesce and Import run details (https://www.servicenow.com/docs/bundle/yokohama-integrate-applications/page/administer/import-sets/concept/c_ImportSetCoalesce.html).